

LAB211 Assignment

Type:	Short Assignment
Code:	J1.S.P0002
LOC:	40
Slot(s):	1

Title

Selection sort algorithm.

Background Context

Selection sort is one of the $O(n^2)$ sorting algorithms, which makes it quite inefficient for sorting large data volumes. Selection sort is notable for its programming simplicity and it can over perform other sorts in certain situations (see complexity analysis for more details).

Program Specifications

Design a program that allows users to input the number of array. Generate random integer in number range input. Display unsorted array and sorted array using selection sort.

Function details:

1. Display a screen to prompt users to input a positive decimal number.
 - Users run the program, display a screen to ask users to enter a positive decimal number.
 - Users input a positive decimal number. Then, perform **Function 2**.
2. Display & sort array.
 - Generate random integer in number range for each array element.
 - Display array before and after sorting.

Expectation of User interface:

```
Enter number of array:
10
Unsorted array: [2, 6, 3, 6, 8, 6, 1, 2, 9, 8]
Sorted array: [1, 2, 2, 3, 6, 6, 6, 8, 8, 9]BUILD SUCCESSFUL (total time: 1 second)
```